Software Engineering — Assignment 2: Agile Sprint Planning & Burndown Analysis

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Student Name	SRN	Section	Course & Assignment	Jira Project Space
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1. Project Epics Cover Four Core Functional Workflows

The **Clinical Trial Recruitment Agent (CTRA)** backlog is structured around four Epics in Jira that divide the clinical recruitment lifecycle into candidate intake and consent, medical eligibility evaluation, researcher compliance protocols, and automated user notifications.

Epic Key	Epic Name	Primary Functional Scope
CTRA-1	Candidate Survey and Consent Filling	Defines upfront eligibility requirements and simplifies candidate intake and informed consent forms.
CTRA-2	Candidature Evaluation and Flagging	Supports de-identified medical screening for researchers and status/flagging transparency for candidates.
CTRA-3	Researcher Compliance and Protocol	Streamlines clinical protocol management and compliance verification workflows for researchers.
CTRA-4	Notification alerts and Connections	Filters and delivers relevant status notifications and communication alerts to platform users.

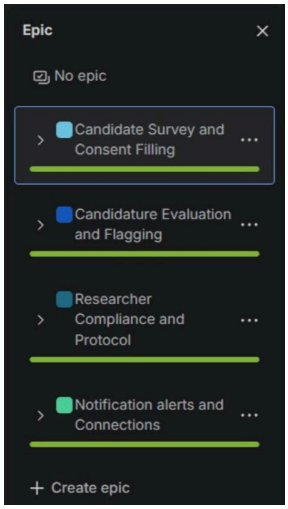


Figure 1: Jira Epic panel for the Clinical Trial Recruitment Agent (CTRA-1 through CTRA-4) showing green completion bars across all four Epics.

2. User Stories Define Role-Specific Requirements and Story Point Estimates

Each Epic is broken down into user stories capturing the stakeholder role, required functionality, and core benefit. Priorities and story point estimates were established to ensure foundational intake and evaluation workflows were completed first.

Issue Key	Parent Epic	User Story Statement	Priority	Story Points	Sprint	Status
CTRA-5	CTRA-1: Candidate Survey and Consent Filling	As an applying candidate, I would like to know the requirements when filling the form so I dont waste time. I would like the form to be simple to understand.	Highest	2	Sprint 2 (+2)	Done
CTRA-6	CTRA-2: Candidature Evaluation and Flagging	As a researcher, I would like to know the details (not personal, just medical) of the chosen candidates. As a candidate, I would like to know if Im selected or flagged and the reason if flagged.	High	1	Sprint 2 (+2)	Done
CTRA-7	CTRA-3: Researcher Compliance and Protocol	As a researcher, I would like the process to be easy.	Medium	5	Sprint 2 (+2)	Done
CTRA-8	CTRA-4: Notification alerts and Connections	As a user, I would only want the relevant notifications.	Medium	1	Sprint 3 (+1)	Done

2.1 Candidate Survey & Eligibility Transparency (CTRA-5 & CTRA-6)

CTRA-5 (Highest Priority, 2 Story Points) addresses candidate onboarding under Epic CTRA-1 by presenting eligibility criteria upfront and keeping the consent form simple. **CTRA-6** (High Priority, 1 Story Point) pairs researcher access to de-identified medical details with transparent candidate selection or flagging reasons under Epic CTRA-2.

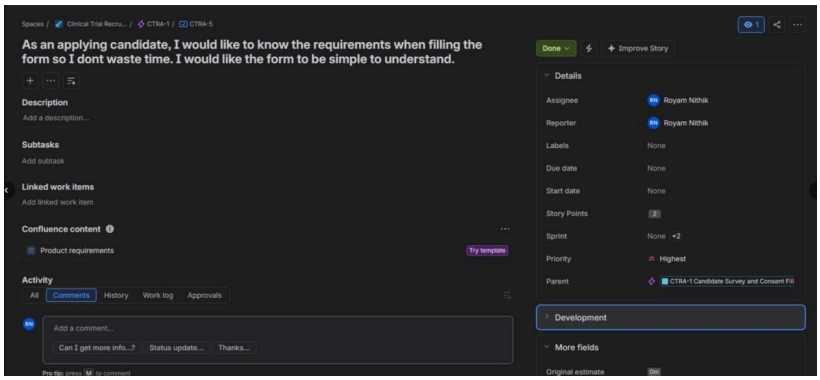


Figure 2: User Story CTRA-5 (Highest Priority, 2 Story Points, Status: Done) — Candidate intake requirements and form simplicity under Epic CTRA-1.

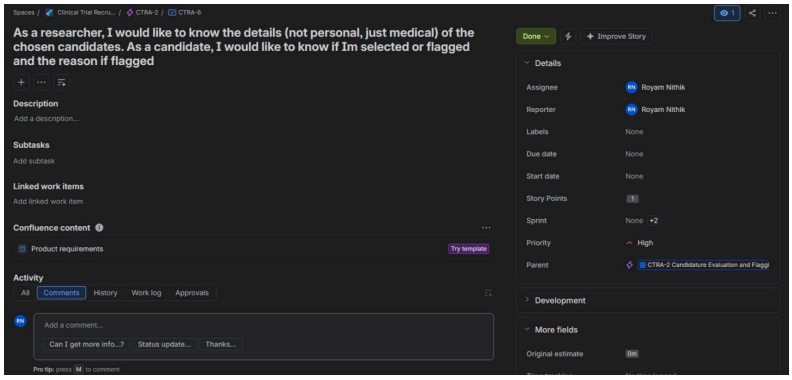


Figure 3: User Story CTRA-6 (High Priority, 1 Story Point, Status: Done) — Non-personal medical details for researchers and flagging feedback for candidates under Epic CTRA-2.

2.2 Researcher Protocol Usability & Relevant Notifications (CTRA-7 & CTRA-8)

CTRA-7 (Medium Priority, 5 Story Points) focuses on simplifying researcher compliance and protocol execution under Epic CTRA-3, representing the largest single effort estimate in the backlog. **CTRA-8** (Medium Priority, 1 Story Point) completes the workflow in Sprint 3 by ensuring users receive only relevant notifications under Epic CTRA-4.

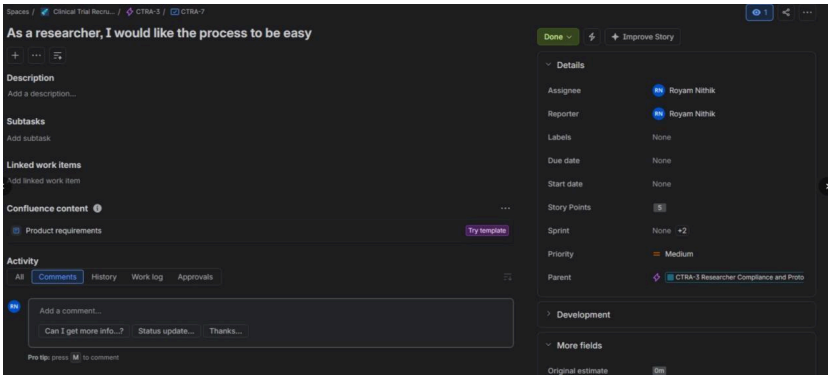


Figure 4: User Story CTRA-7 (Medium Priority, 5 Story Points, Status: Done) — Streamlined researcher compliance and protocol workflow under Epic CTRA-3.

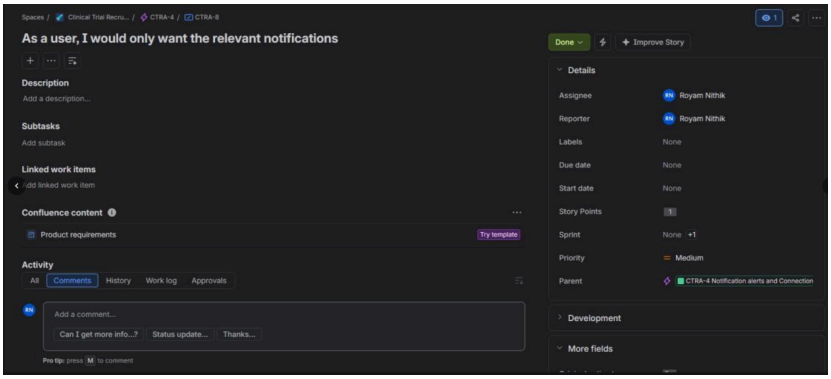


Figure 5: User Story CTRA-8 (Medium Priority, 1 Story Point, Status: Done) — Targeted, relevant notification delivery under Epic CTRA-4.

3. Sprint Burndown Charts Track Issue Completion Across Working and Non-Working Days

The Jira Burndown Charts track remaining **Issue Count** from August 26 to September 1 (with August 29–30 shaded as non-working days). **CTRA Sprint 2** tracked 3 issues to completion, while **CTRA Sprint 3** tracked the final notification issue to completion.

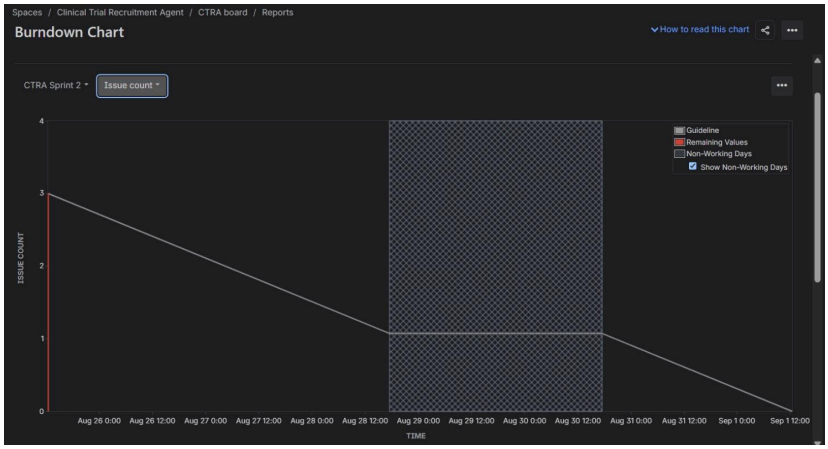


Figure 6: CTRA Sprint 2 Burndown Chart (Issue Count: 3 issues) showing progress across August 26 – September 1 with full completion by sprint close.

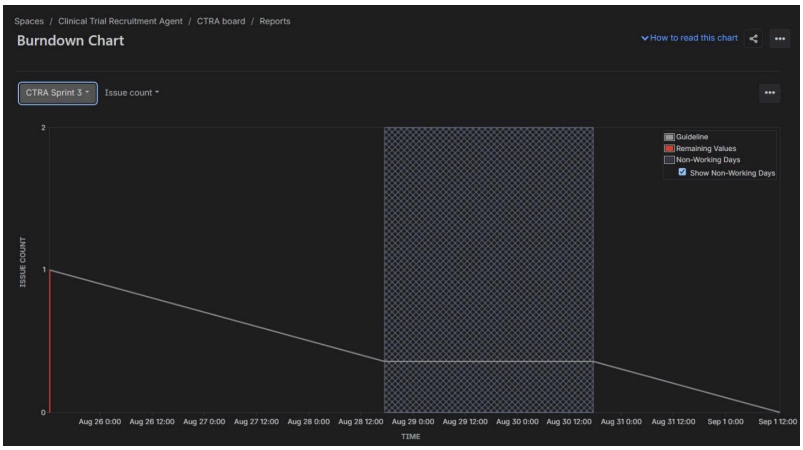


Figure 7: CTRA Sprint 3 Burndown Chart (Issue Count: 1 issue) showing completion of the remaining backlog item by September 1.

4. Sprint Reflection Questions & Responses

The following responses address the four sprint reflection questions on estimation accuracy, backlog prioritization, sprint plan alignment, and team capacity insights from the burndown charts.

#	Reflection Question	Response
1	**Did your estimations reflect the actual effort?**	No, as we were made more aware of the needs while working on them.
2	**Was your backlog well-prioritized?**	Yes, we made sure to prioritize more integral and important parts that other functionalities depend on first.
3	**How did your simulated sprint align with your plan?**	We tried adhering to the original plan and failed. We took longer the more we got familiar with the problem.
4	**What insights did the burndown chart give about your team's capacity?**	That though we were behind, we were able to finish all the issues.